Homework 12
Introduction to Complex Analysis
(MATH 4230-40), Spring 2022

Section 4.4 - Evaluation of Infinite Series and Partial-Fraction Expansions

1. Show that $\displaystyle\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{(2n-1)^3} = \frac{\pi^3}{32}$. You may assume that $\displaystyle\lim_{N\to\infty} \int_{C_N} \left(\frac{\pi}{\sin(\pi z)}\right)\left(\frac{1}{(2n-1)^3}\right) dz = 0$, where C_N is the square with corners $\pm(N+1/3) \pm (N+1/3)i$.

Section 5.1 - Basic Theory of Conformal Mappings

2. Near what points are the following maps conformal?
   (a) $f(z) = \overline{z}$
   (b) $f(z) = \dfrac{\sin z}{\cos z}$
3. If $f : A \to B$ is bijective and analytic with an analytic inverse, prove that f is conformal.